**AERO60492 – Autonomous Mobile Robots**

## Answers

The following questions challenge you to read documentation for several services which communicate using a serial protocol and answer questions about their meaning and use.

## Example 1: Windmaster

Using the "windmaster_manual.pdf" (which can be found on Blackboard), answer the following questions:

1. What does this device do?

An internet search of the "Ultrasonic Anemometer" should reveal this device measured wind-speeds. Also, throughout the manual "wind speed" is mentioned very often. Page 4 has these lines:

> "The WindMaster, constructed in aluminium and carbon fibre, can perform measurements in wind speeds up to 45 m/s.
>
> The WindMaster Pro, constructed from marine grade stainless steel 316, can perform measurements in wind speeds up to 65 m/s."

2. How many serial data frame output options are there?

Section 7 (page 26) of the manual provides an overview of the serial message format options. The 'message format' is synonymous with the 'frame specification'. There are 8 formats listed – Mode 1 2 3 4 7 8 9 10.

3. For the 'Binary, Polar, Short' mode, how many bytes are in each data frame?

From the mode overview, we see Binary Polar Short is listed as Mode 7 (detailed on page 31). The frame specification is:

<B1><B1>STATUS,DIR,MAG,W,SOS,CHECKSUM

Just below we see that "Data is 2 byte signed two's complement (except unsigned for SOS; 1 byte for checksum)." The <B1> is used to represent 1 byte. Therefore, we have:

| Item: | <B1> | <B1> | STATUS | DIR | MAG | W | SOS | CHECKSUM | |
|---|---|---|---|---|---|---|---|---|---|
| Bytes: | 1 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | Total: 13 bytes |

4. What is status code '0A' mean?

From the contents page, section 11.4 details the "Status (error) codes". Looking at page 47 reveals '0A' represents "System gain at Maximum".

5. What are the default and maximum frame rates?

Searching the document for the keyword “rate” shows a table in section 4. Specification. There is a line listing the “Output Rate” options. Listed are: 1, 2, 4, 8, 10, 16, 20, (32 option) Hz. So maximum is 32 samples/frames per second.

Further searching against the keyword “default” shows the output is set as 1 Hz on page 27:

“The WindMaster delivers wind information at rates from 1 (default setting) to 32 outputs / second”

6. What is the type of checksum used?

The checksum is identical for all serial message formats but only specified once for the Mode 1 on page 27.

> Checksum
>
> This enables the host system to check that the data has been correctly received. This is the EXCLUSIVE – OR of the bytes between (and not including) the and characters reported as ASCII Hex.

## Example 2: MAVLink

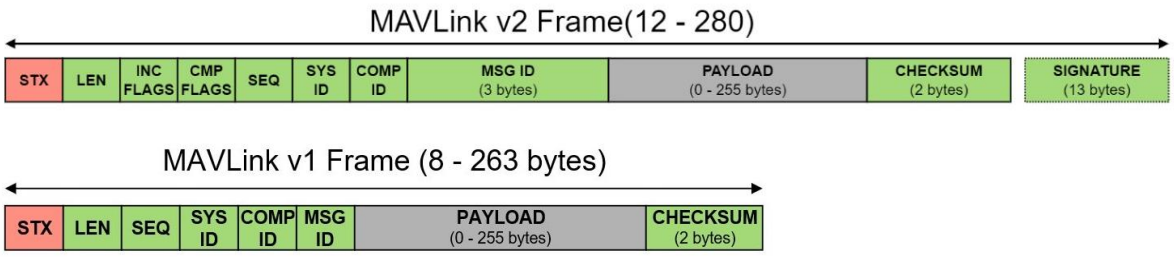
Using the MAVLink serialisation guide (https://mavlink.io/en/guide/serialization.html), answer the following questions:

1. What does the system ID represent?

From the table: “ID of system (vehicle) sending the message. Used to differentiate systems on network. Note that the broadcast address 0 may not be used in this field as it is an invalid source address.”

2. What is the shortest possible frame size in bytes?

Above the serial frame speciation diagrams are the minimum lengths for the MAVLink V1 (8 bytes) and MAVLink V2 (12 bytes).




3. What is the largest possible frame size in bytes?

Similarly for Q2, the maximum lengths for the MAVLink V1 (263 bytes) and MAVLink V2 (280 bytes).

4. What is MessageID = 28 for?

We see the "MSG ID" is described as "ID of *message type* in payload. Used to decode data back into message object." Therefore, its used to differentiate between a selection of different payloads.

An internet search for "MAVLink message ID 28" finds this page:

https://mavlink.io/en/messages/common.html

Scrolling to #28 shows its "RAW_PRESSURE (28)"

5. Is little or big endian used?

Near the top of the MAVLink specification page there is a statement:

> "Note that multi-byte fields are serialized in little-endian format,"

6. What is the frame start byte/character?

From the frame specification table, the start bytes (termed 'STX') are described to have value 0xFD. This would appear as the 'ý' if the bytes is interpreted using the extended ASCII encoding[1].

## Example 3: NMEA Messages

Using "NMEA Reference Manual-Rev2.1-Dec07.pdf" (which can be found on Blackboard), answer the following questions:

1. What system are these messages associated with?

The NMEA message format is associated with Global Navigation Satellite Systems (GNSS).

2. Are the messages binary or plain-text?

The manual provides example serial messages for each of the various message types supported by GNSS systems. For example, the 'GGA' message has the example:

> $GPGGA,002153.000,3342.6618,N,11751.3858,W,1,10,1.2,27.0,M,-34.2,M,,0000*5E

Since all the character can be interpreted directly (i.e its possible to see the Northing coordinate is 3342.6618) then this is a plain text format.

[1] https://byte-tools.com/en/ascii/

3. Using this data frame:
   $GPGGA,002153.000,5248.65262,N,00407.44342,W,1,10,1.2,27.0,M,-34.2,M,,0000*5F
   (a) What is at this location?

We see this is a 'GGA' type message and is defined more fully on page 1-2. The location is specified by the latitude (North/South) and Longitude (East/West) fields, with the formats:

Latitude: ddmm.mmmm

Longitude: dddmm.mmmm

Therefore, the numerical values need some minor decoding to assign the correct unit to each value. We start with the number of degrees which is the first two (52 deg) and three numbers (004 deg) for lat and long respectively. The next two values and the values after the decimal are the minutes specified as a floating-point value. Hence the location is:

52°48.65262'N 4°7.44342'W

Using an online map service (note not all search engines accept this format so a conversion might be needed[2]) shows this location is the "RAF Llanbedr" (on Google) or the Snowdonia Aerospace Centre.

(b) Is the checksum correct? (requires a little code)

Searching the NMEA specification document reveals the definition of the checksum:

> "The field of checksum consists of two hex digits representing the exclusive or of all characters between, but not including, the $ and *"

It is the two values appearing after the asterisk *. For this example, the check sum given is 5F. Therefore, we need to take the byte/hex value of each character in the NMEA string, and perform the "exclusive or" operation. There are fortunately handy tools[3] to perform this action. The checksum should be 5C, hence the checksum is incorrect.

**Example 4: GasCard NG**

Using the "Procedures-for-NGA-RS232-communicationv1_03.pdf" (which can be found on Blackboard), answer the following questions:

1. What does this device do?

There is little information on the actual purpose inside the manual. An internet search should reveal this is a device measures gas concentration.

[2] https://www.eddmaps.org/tools/latlongconv.cfm
[3] https://www.meme.au/nmea-checksum.html

2. In 'Normal Mode', is the serial output binary or plain text?

Reading section "2.2. Normal Mode" shows the serial frame specification with an example. Fields such as concentration and air pressure are shown as numerical values. Since they can be direct interpreted without the need to convert from bytes, this is a plain text format.

There is also a statement on page 2:

> "Data from the Gascard NG is sent as ASCII with spaces between the fields"

3. How many bytes per data frame?

On page 2, there is an example serial frame specification:

N<Space>0.3647<Space>-0.0000<Space>-0.0000<Space>-0.0000<Space>-0<Space>33265<Space>1071.8<Space>0<CR><LF>

Each item between <> character represents one byte. Each letter, number, or – (minus) symbol is also one byte. Therefore, counting along the entire frame (ignoring the <> as they are just to aid interpretation), the frame length is: 52 character (hence 52 bytes) for this message format.

4. What different units are used within each frame?

This is a tricky question to answer and requires somewhat in-depth understanding of the GASCARD system, however we have some hints from the manual.

- Pressure is defined in millibar and the example frame values of "983.1" are in line with standard atmospheric conditions.
- The temperature is shown as 30000. This is likely to represent 30.000 degrees since 300 or 3 degrees seems unusual for a normal example.
- The concentrations values are described as "Conc 1 is the fraction of the full scale concentration currently being indicated.". Since the GASCARD can be configured for multiple gas species, this is a fractional value of whatever unit the maximum sensing ability is.

5. What is the hardware serial transmission protocol used?

The manual shows a serial protocol specification of 576008N1 which is associated with several hardware layers. In this case the cover page of the manual makes clear reference to the RS232 protocol.